Jan 2019

Q2 Differentiation & Integration

a) $y = x^3 - 12x + 25$ A = Max turning point, C = Min turning point

i) $\frac{dy}{dx} = 3x^2 - 12$

| $y = x^n$ | power Rule |
|---|---|
| $\frac{dy}{dx} = nx^{n-1}$ | |

$\frac{d^2y}{dx^2} = 6x$

ii) at point B, $x = 0$.

sub $x = 0$ into $dy/dx$

$\therefore$ slope $= 3(0)^2 - 12$

$= \boxed{-12}$

iii) let $dy/dx = 0$ and solve for $x$.

$3x^2 - 12 = 0$

$3x^2 = 12$

$x^2 = 12/3 = 4$

$x = +2$ or $x = -2$

when $x = 2$: $y = 2^3 - 12(2) + 25$

$= 9$

C: $(2, 9)$

$x = -2$: $y = -2^3 - 12(-2) + 25$

$= -8 + 24 + 25$

$= 41$

A: $(-2, 41)$

b) i) $y = \frac{3x}{2x+1}$ ($3x$: $u$; $2x+1$: $v$)

use Quotient Rule:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$u = 3x \quad \frac{du}{dx} = 3$

$v = 2x+1 \quad \frac{dv}{dx} = 2$

sub into the formula.

$$\frac{df}{dx} = \frac{(2x+1)(3) - (3x)(2)}{(2x+1)^2}$$

$$= \frac{6x+3-6x}{(2x+1)^2}$$

$$= \boxed{\frac{3}{(2x+1)^2}}$$

ii) $y = (4x^2+9)^3$ ($4x^2+9$: $u$)

use Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$u = 4x^2+9 \quad \frac{du}{dx} = 8x$

$y = u^3 \quad \frac{dy}{du} = 3u^2$

sub into the formula.

$$\frac{dy}{dx} = (3u^2).(8x)$$

$$= 3(4x^2+9)^2(8x)$$

$$= 24x(4x^2+9)^2$$

c) $I = 1 + 30t - 5t^2$

i) current 1 second after activation : $t = 1$

$I = 1 + 30(1) - 5(1)^2$

$= 26$ milliamps

ii) rate at which current changing after $t = 2$

get $\frac{dI}{dt} = 30 - 10t$ sub in $t = 2$

$\Rightarrow 30 - 10(2) = 10$ milliamps / second

iii) time take to reach first peak (i.e. turning point)

$\downarrow$

$dy/dx = 0$.

let $\frac{dI}{dt} = 0$ and solve for $t$

$30 - 10t = 0$

$10t = 30$

$t = 3$ seconds.

d) area enclosed by the curve $y = x + 2x^2$ $x = 0$ & $x = 2$

$$\int_0^2 (x + 2x^2)\,dx = \frac{x^2}{2} + \frac{2x^3}{3}\Big]_0^2 \quad \text{(definite integral)}$$

$$= \left(\frac{2^2}{2} + \frac{2(2)^3}{3}\right) - \left(\frac{0^2}{2} + \frac{2(0)^3}{3}\right)$$

$= \frac{22}{3}$ or $7.33$ square units.